Rosalind Franklin was a British chemist whose revelatory work was vital to the discovery of the structure of DNA. Yet it was largely overlooked in her lifetime.

Born in London in 1920, Rosalind Franklin graduated from the University of Cambridge in 1941 with a degree in physical chemistry. She then took a research post studying the chemical structure of coal and graphite before gaining her doctorate from Cambridge in 1945. Moving to Paris in 1947, Franklin studied X-ray crystallography—using X-rays to examine the crystalline form of a substance and to determine its molecular structure. It was her expertise in this field that enabled her to make a crucial contribution to one of the biggest scientific conundrums of the 20th century.

The DNA race

During the 1950s, a number of scientists across the globe were engaged in a “race” to discover the molecular structure of deoxyribonucleic acid (DNA), the chemical found inside all living things, which encodes genetic information. In Cambridge in 1951, James Watson and Francis Crick were attempting to

In 1951, Franklin began photographing *strands of DNA at King’s College, London, where a group of physicists, biologists, and biochemists were helping pioneer biophysics (using physics to study biological molecules).*

MILESTONES

CHEMICAL RESEARCH
Becomes a researcher for BCURA, the British Coal Utilization Research Association, in 1942.

NEW CAREER
Trains under French engineer Jacques Mering in 1947 and becomes an X-ray crystallographer.

KING’S COLLEGE, LONDON
Asked to head X-ray research in 1951, without the knowledge of post-holder Maurice Wilkins.

DNA BREAKTHROUGH
Produces the first sharp image of crystalline DNA in 1952, which reveals its double helix shape.

PIONEERING WORK
Moves to Birkbeck College, London, in 1953 and leads groundbreaking studies into major crop viruses.

“Science and **everyday life cannot** and **should not be separated.”**

Rosalind Franklin, 1940